

Chap 1 (Felix Munoz Garcia)

What is preference-based approach of consumer's behavior?

Explain and interpret the assumption of preference based approach (e.g., Completeness, Reflexivity, Rationality)

Ans: There are two ways to approach the decision making process of a consumer, "preference-based approach" and "choice-based approach".

"Preference-based approach" analyzes how the individual uses his preferences to choose an element(s) from the set of alternative X . This approach contrasts from the "choice-based approach" where actual choices of the individual is analyzed.

Even though preference-based approach is based on unobservables, it is more tractable when the set of alternative X has many elements.

In the preference based approach we understand the decision maker's preferences as a his attitude towards the elements in a set of possible alternatives X .

These attitude could be obtained by presenting a questionnaire to the individuals like the following one:

For any $x, y \in X$ how do you compare x and y ?

- ☐ prefer x to y ($x \succ y$)
- ☐ prefer y to x ($y \succ x$)
- ☐ am indifferent ($x \sim y$)

Assumptions of preference-based approach:

Completeness: A preference relation is complete if, for any two alternatives $x, y \in X$, either: alternative x is preferred to y , $x \succ y$ or alternative y is preferred to x , $y \succ x$ or the individual is indifferent between x and y , $x \sim y$.

The completeness assumption suggests that the individual is capable of comparing any pair of alternatives that we present to him.

A remark about completeness is, not all binary relations satisfy completeness.

Reflexivity: A preference relation satisfies

reflexivity if for any alternative

x in X , we have that $x \succeq x$,

so that any bundle is weakly preferred to itself.

This property entails that

i) $x \sim x$, so that any bundle is indifferent to itself, and

ii) $x \not\prec x$

This assumption ensures that any

bundle belongs to at least one

indifference set, namely the set

containing itself if nothing else

However, reflexivity is implied from completeness.

Rationality: We say a preference relation $\succeq$ is rational if it satisfies the following two properties:

Completeness: For any pair of alternatives $x, y \in X$, either $x \succeq y$, or $y \succeq x$, or both $(x \sim y)$.

Transitivity: For any three alternatives $x, y, z \in X$, if $x \succeq y$ and $y \succeq z$, then it must be that $x \succeq z$.

Example: Consider the preference relation $\succeq$ defined as $x \succeq y$ if and only if $\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i$.

In words, the consumer prefers bundle x to y if the total number of goods in bundle x is larger than in bundle y .

The preference relation is complete since for any two bundles $x, y \in \mathbb{R}^N$, either $x \succeq y$ because $\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i$, or $y \succeq x$ because $\sum_{i=1}^N y_i \geq \sum_{i=1}^N x_i$, or both because $\sum_{i=1}^N x_i = \sum_{i=1}^N y_i$.

It is also transitive because if $x \succeq y$ and $y \succeq z$, then

$$\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i \geq \sum_{i=1}^N z_i$$

which entails that $\sum_{i=1}^N x_i \geq \sum_{i=1}^N z_i$, thus

implying $x \succeq z$, as required for transitivity.

2. Define utility function.
Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity.

Ans: A function $u: X \rightarrow \mathbb{R}$ from the

Utility function: A function $u: X \rightarrow \mathbb{R}$ from the set of alternatives to the set of real numbers is a utility function representing a preference relation $\succeq$ if, for every pair of alternatives $x, y \in X$, $x \succeq y$ is equivalent to $u(x) \geq u(y)$

We can emphasize two points in this definition.

First, only the ranking of alternative matters. Individual does not care about cardinality but instead cares about ordinality.

Second, If we apply any strictly increasing function $f(\cdot)$ on the utility function $u(x)$, that is, $v(x) = f(u(x))$, the new function keeps the ranking of alternatives intact, and therefore the new function still represents the same preference relation.

Explaining the desirability of a consumer's preference by the axiom of monotonicity and strong monotonicity.

In order to do that, let us first consider x is an N -dimensional bundle, $x \in \mathbb{R}^N$, that is $x = (x_1, x_2, \dots, x_N)$, where its k th component represents the amount of good (or service) k , $x_k \in \mathbb{R}$.

Monotonicity: A preference relation satisfies monotonicity if, for any two bundles $x, y \in X$, where $x \neq y$,

$x_k \geq y_k$ for every good k , implies $x \succeq y$

and $x_k > y_k$ for every good k , implies $x \succ y$

That is, increasing the amount of some commodities (without reducing the amount of any other commodity) can not hurt, $x \succeq y$, and increasing the amount of all commodities is strictly preferred, $x \succ y$.

Strong monotonicity: A preference relation satisfies strong monotonicity if for any two bundles $x, y \in X$, where $x \neq y$, and $x_k \geq y_k$ for every good k , we have that $x \succ y$.

That is, even if we increase the amounts of only at one of the commodities, we make the consumer strictly better off.

3. Show that

Strong monotonicity $\iff$ Monotonicity

Ans: There is a connection between preference relations and the utility function representing them.

Monotonicity in preferences implies that the utility function is weakly increasing in its arguments. That is, increasing some of its arguments weakly increases the value of the utility function, and increasing all its arguments strictly increases its value.

More formally, for a common factor $\alpha > 1$, monotonicity in preferences implies that

$$u(\alpha x_1, x_2) \geq u(x_1, x_2) \quad \text{and}$$

$$u(\alpha x_1, \alpha x_2) > u(x_1, x_2)$$

However, strong monotonicity is preferences implies that the utility function is strictly monotone (strictly increasing) in all its arguments. That is, increasing some of its arguments strictly increases the values of the utility function.

More formally strong monotonicity in preferences implies that for $\alpha > 1$

$$u(\alpha x_1, x_2) > u(x_1, x_2)$$

Now we consider two following examples.

Example 1: $u(x_1, x_2) = \min\{ax_1, bx_2\}$

where $a, b > 0$, representing that goods 1 and 2 are regarded as perfect complements.

The preference relation represented by the above utility function satisfies monotonicity.

Because, if we increase all arguments by a common factor $\delta > 0$, we obtain

$$\min\{ax_1 + \delta, bx_2 + \delta\} > \min\{ax_1, bx_2\}$$

The preference relation represented by the above utility function satisfies monotonicity.

Because if we increase all arguments by a common factor $\delta > 0$, we obtain

However, this result does not satisfy strong monotonicity. Because, if we increase one argument alone (e.g., amount of good 1), we obtain

$$\min \{ax_1 + \delta, bx_2\} \not\geq \min \{ax_1, bx_2\}$$

for a case where $\min \{ax_1, bx_2\} = bx_2$

Example 2: $u(x_1, x_2) = ax_1 + bx_2$
 where $a, b > 0$, which denotes that individual regards good 1 and good 2 as perfect substitutes.

The preference relation represented by the above utility function satisfies monotonicity. Because if we increase all arguments,

we obtain,

$$(ax_1 + \delta) + (bx_2 + \delta) > ax_1 + bx_2$$

for all $\delta > 0$

It also satisfies strong monotonicity,

since, if we increase one argument alone (e.g., the amount of good 1),

we obtain:

$$(ax_1 + \delta) + bx_2 > ax_1 + bx_2$$

~~Combining our results~~

Combining the results of

above two examples, we can see that

a preference relation that satisfies

strong monotonicity, such as that in

example 2, must also satisfy monotonicity

But the opposite is not necessarily true. A preference relation that satisfies monotonicity, such as that in example 1, does not necessarily satisfy

strong monotonicity

Strong monotonicity $\iff$ monotonicity

4. Show that

Strong monotonicity $\Leftrightarrow$ Monotonicity $\Leftrightarrow$ LNS

Ans: Other than monotonicity and strong monotonicity, desirability of consumer's preference can be expressed through non-satiation (NS) and local non-satiation (LNS).

A preference relation satisfies non-satiation (NS) if, for every bundle $x \in X$, we can find another bundle $y \in X$ that is strictly preferred to x , i.e., $y \succ x$.

But the notion of non-satiation is too general, since we could think about a bundle y containing extremely larger amounts of some goods than x .

Therefore, we need to look for a more restricted version, local non-satiation (LNS).

A preference relation satisfies local non-satiation (LNS) if for every bundle $x \in X$, we can find another bundle $y \in X$ that is strictly preferred to x , $y \succ x$, and where y is also ε -away from x , $\|y - x\| < \varepsilon$ for every $\varepsilon > 0$.

Monotonicity and LNS: If a preference relation satisfies monotonicity, it must also satisfy LNS.

Let, a bundle $x = (x_1, x_2)$

If we increase all of its components, we obtain $(x_1 + \delta, x_2 + \delta)$ that is strictly preferred to bundle (x_1, x_2) by the definition of monotonicity.

Hence there is a bundle

$$y = (x_1 + \delta, x_2 + \delta)$$

Hence there is a bundle

$$y = (x_1 + \delta, x_2 + \delta) \text{ such that } y \succ x$$

and $\|y - x\| < \varepsilon$, where δ can be made arbitrarily small, $\delta \rightarrow \varepsilon > 0$.

Therefore monotonic preference relations also satisfy LMS.

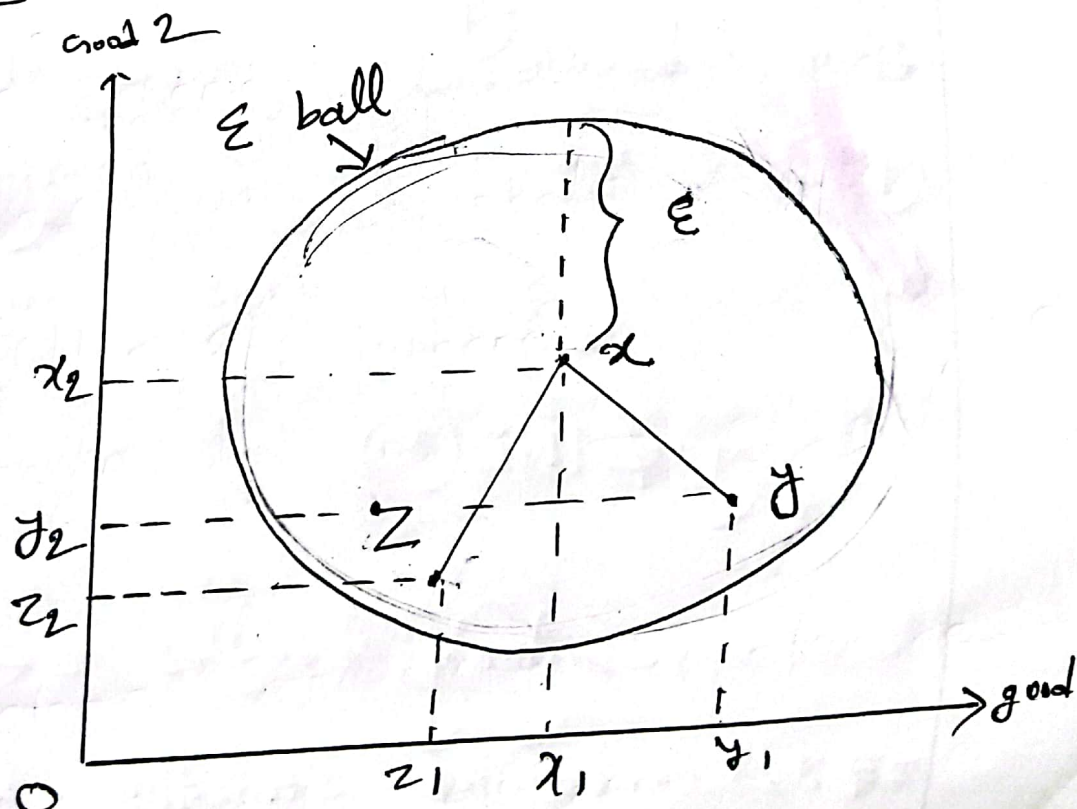
In addition, since LMS is strongly
monotonic preference relations
are also monotonic, strongly
monotonic preference relations
also satisfy LMS .

But the opposite is
not necessarily true. That is, LMS
does not satisfy monotonicity or
strong monotonicity.

Since $\|y - x\|$ represents the
Euclidean distance between x and
 y , there may be such a case
where the individual decision maker
prefers bundle y to x even if
bundle y contains less of good

2 (but more of good 1) than bundle x .

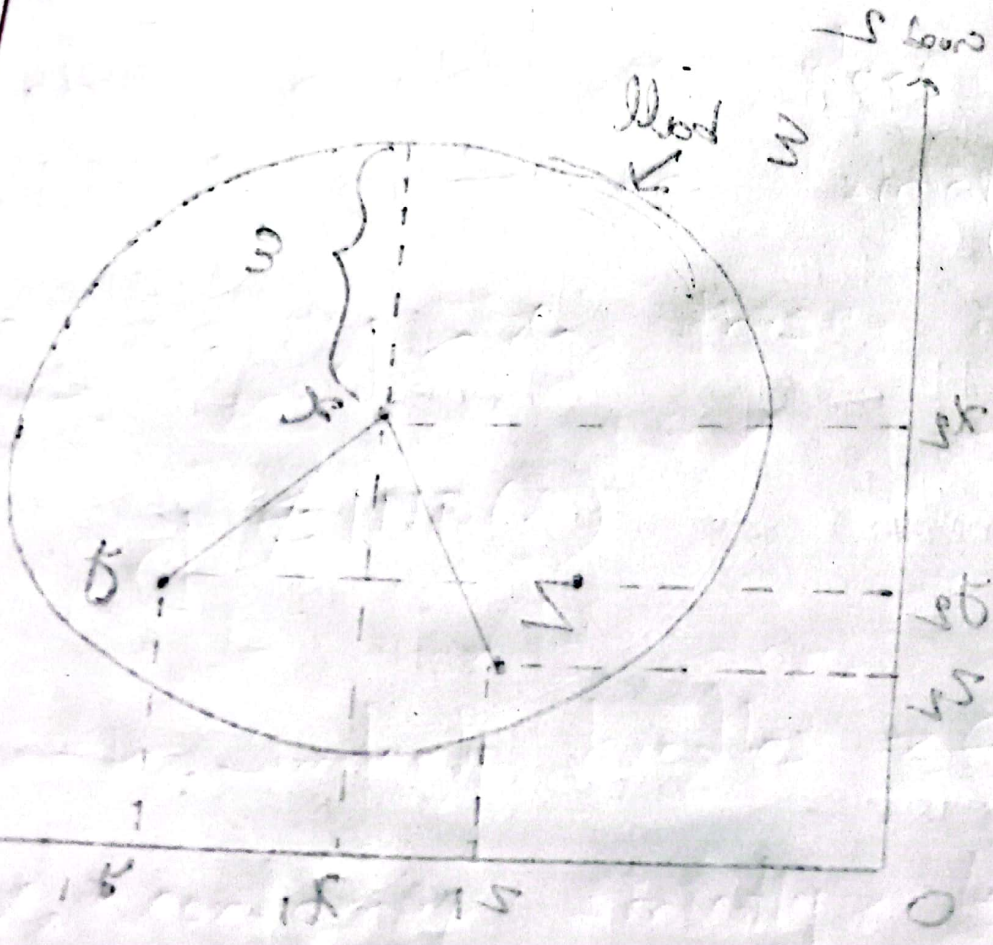
LNS also allows for the possibility that an individual decision maker with an initial bundle x might prefer a bundle z that contains less of both good 1 and 2 than x does. These possibilities are depicted in the following figure



Therefore, LNS does not satisfy monotonicity or strong monotonicity

In summary

Strong monotonicity $\Rightarrow$ Monotonicity $\Rightarrow$ LNS



5. Define indifference set

Explain how does strong monotonicity implies downward sloping indifference curve

Ans: Indifference set. The indifference

set of a bundle $x \in X$ is the set of all bundles $y \in X$ such that $y \sim x$.

More formally, we can write the indifference set of bundle x as

$$Ind(x) = \{y \in X : y \sim x\}$$

Strong monotonicity and indifference curve:

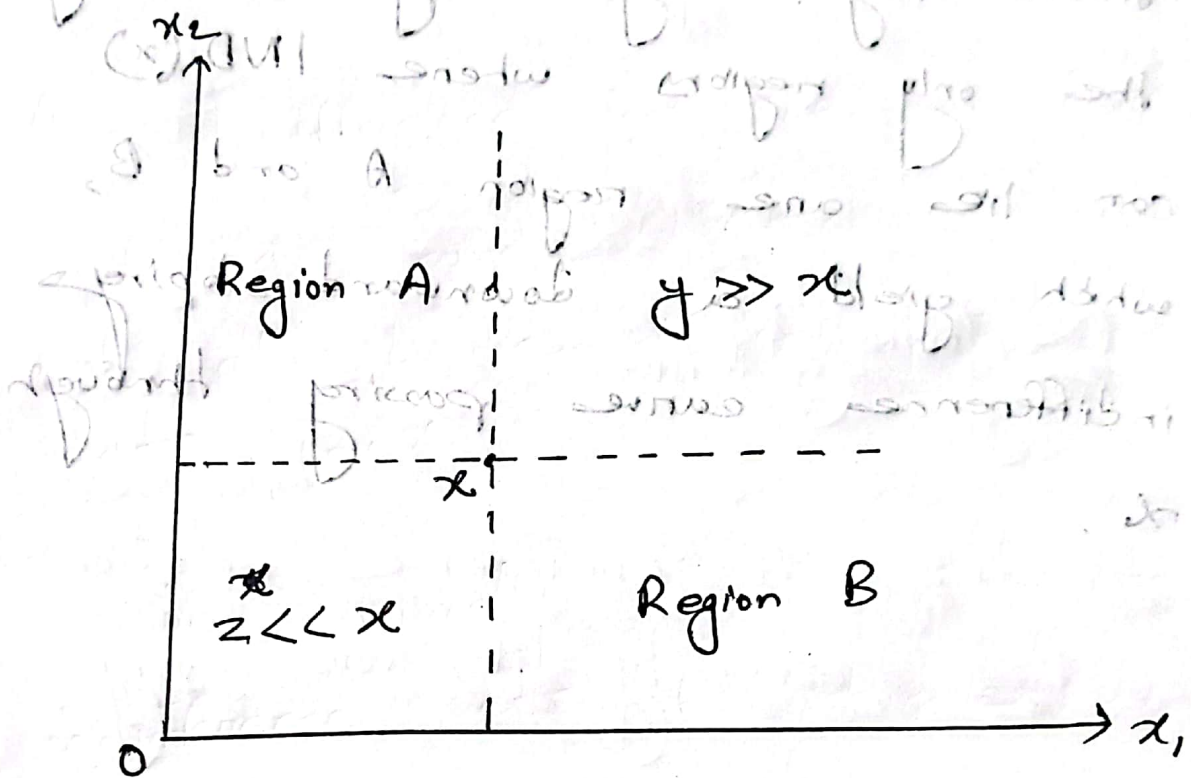
Let consider a bundle $x \in X$

By strong monotonicity, other bundles $y \in X$, where $y \neq x$, that contain strictly more of all goods than bundle x does, that is

$y \gg x$, must be strictly preferred to x , $y \succ x$, thus implying that $y \notin \text{IND}(x)$

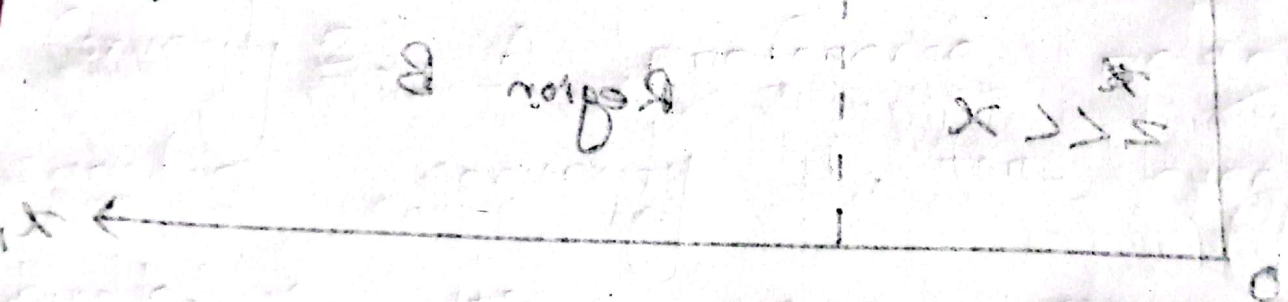
Similarly bundles $z \in X$, where $z \neq x$, containing strictly less of every good than bundle x does,

that is, $z \ll x$, must satisfy $z \ll x$,
 thus entailing that $z \notin \text{IND}(x)$!!



As depicted in the above figure, for
 a given bundle $x \in \mathbb{R}_+^2$, bundles satisfying
 $y \gg x$ lie to the northeast of x
 and bundles satisfying $z \ll x$ lie to the
 southwest of x .

Since the indifference set $IND(x)$ cannot lie on either of these regions by strong monotonicity the only regions where $IND(x)$ can lie are region A and B, which yield a downward sloping indifference curve passing through x .



As indicated in the above figure, given bundle x either belongs to the northern or southern set of all $x > x$ lie to the north of x and all $x < x$ lie to the south of x .

6. What are the twin definitions of convexity?
 In what respect they are different from strict convexity, examine graphically.

Ans:

Twin definition of convexity:

Convexity 1: A preference relation satisfies convexity if, for any two bundles $x, y \in X$, $x \succeq y$ implies that for any $\alpha \in (0, 1)$, the convex combination $\alpha x + (1-\alpha)y$ is also preferred to y .

Intuitively, if bundle x is weakly preferred to bundle y , then any mixture of x and y is also weakly preferred to y .

convex

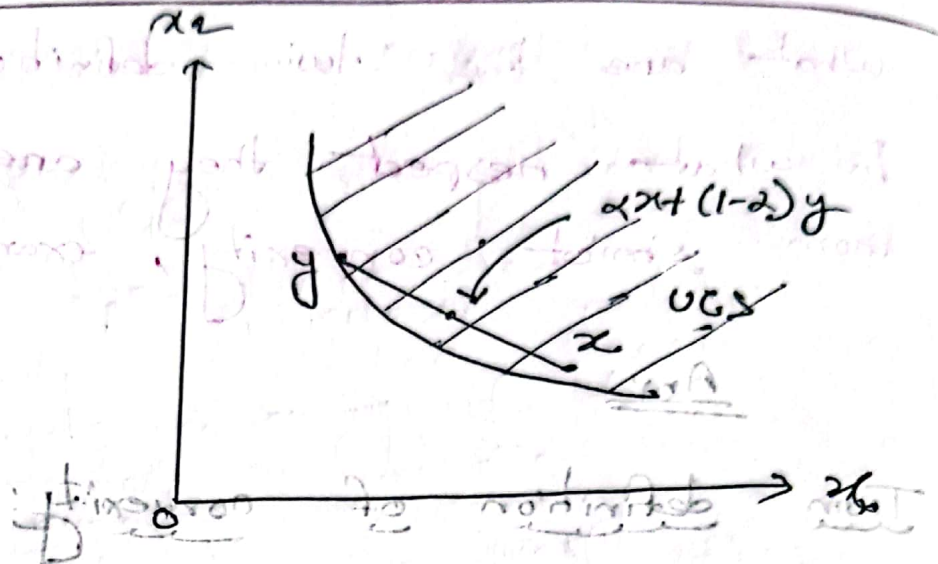


Fig: Convexity

Convexity 2: A preference relation satisfies convexity if, for every bundle x , its upper contour set, $U(x) = \{y \in X : y \succeq x\}$, is convex.

In words, for every two bundles y and z weakly preferred to x , $y \succeq x$ and $z \succeq x$, then convex combination $\alpha y + (1-\alpha)z$ is

also weakly preferred to x .

$$\alpha y + (1-\alpha)z \succeq x \quad \text{for any } \alpha \in (0,1)$$

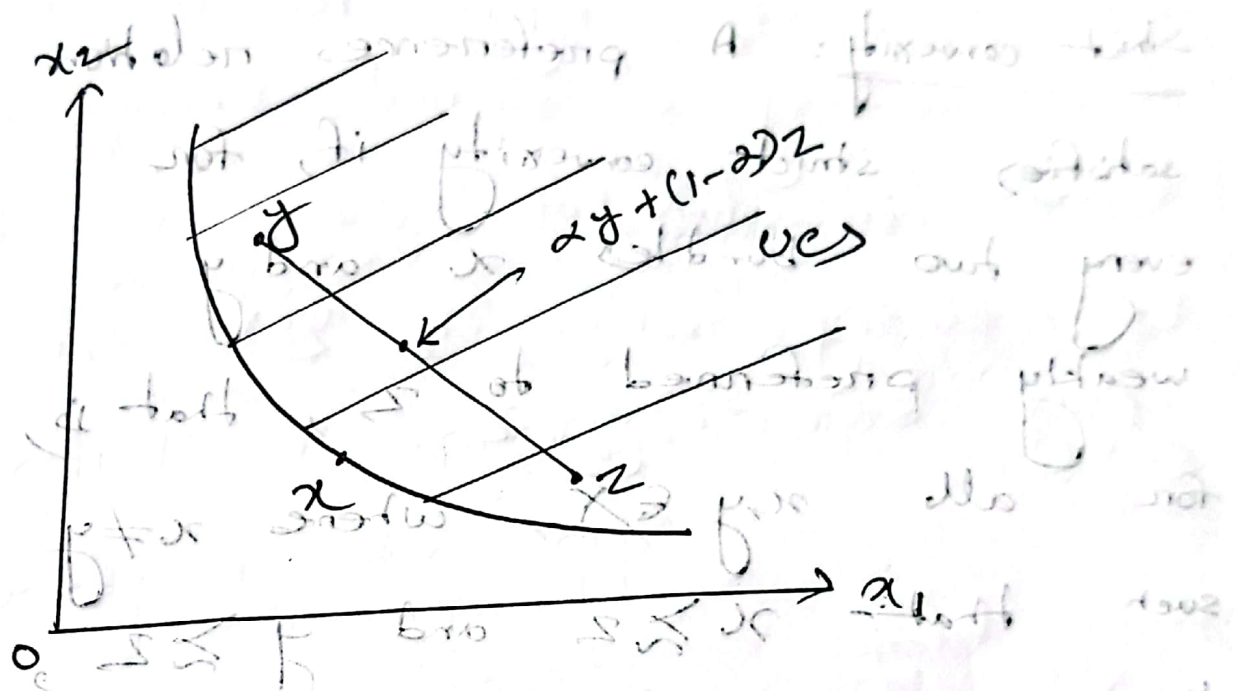


Fig: convexity 2

Graphically, convexity in the preference relation implies that the

indifference curves are bowed in toward the origin.

Difference between convexity and

(1) strict convexity: $x \leq \lambda (x - 1) + \lambda y$

Strict convexity: A preference relation satisfies strict convexity if, for every two bundles x and y weakly preferred to z , that is, for all $x, y \in X$, where $x \neq y$ such that $x \succeq z$ and $y \succeq z$, their convex combination $\lambda x + (1 - \lambda)y$, where $\lambda \in (0, 1)$, is strictly preferred to bundle z .

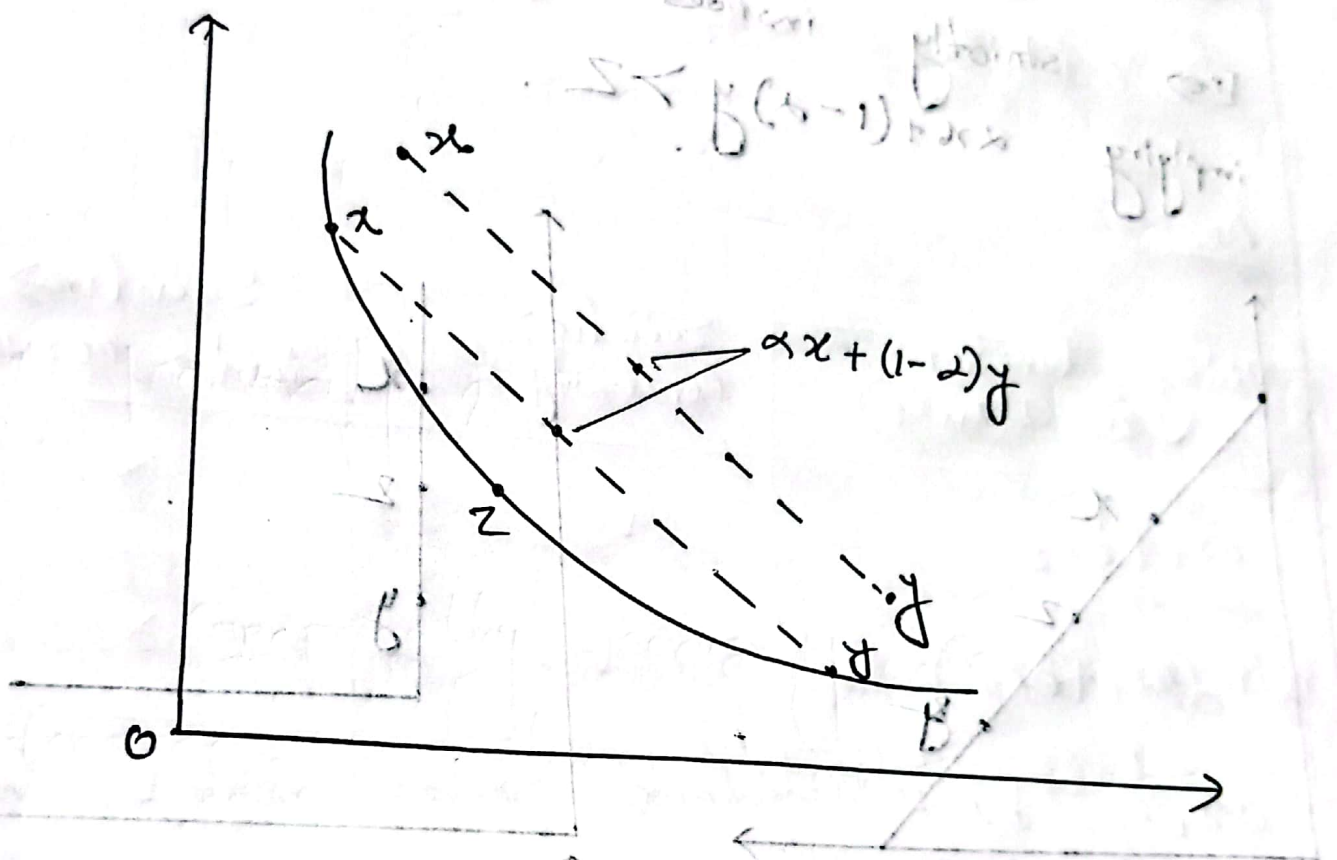
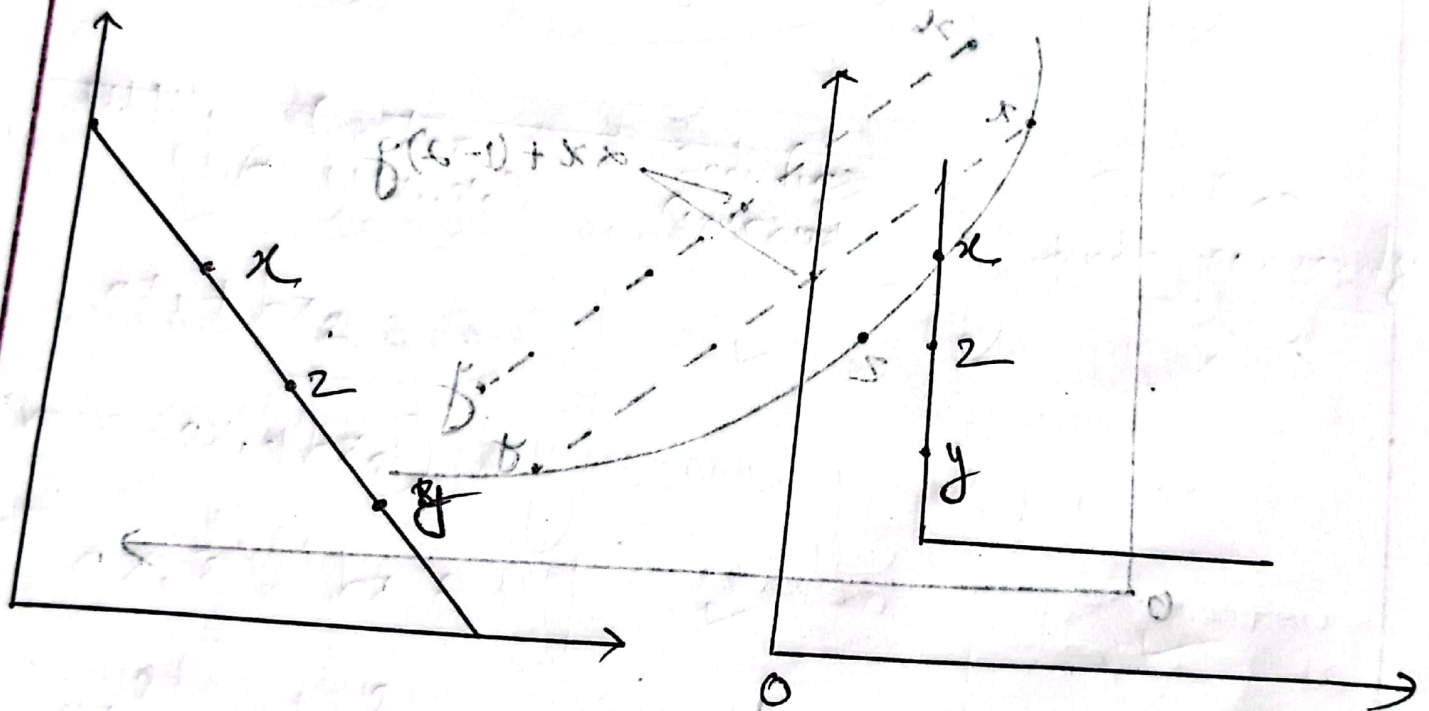


Figure: Strict convexity

Even if bundles x and y lie on both the U_C and U_D indifference curves like the above figure, the definition of strict convexity is not satisfied. Because the convex combination of these two bundles lies

strictly inside the U_C , thus implying $\alpha x + (1-\alpha)y > z$.



(a)

(b)

Fig: Convexity but not strict convexity

The above figure illustrates a preference relation that satisfies convexity but not strict convexity: Because the convex combination of bundle x and y produces a bundle that does not lie strictly within the UC

$$\alpha x + (1-\alpha)y \succeq z, \text{ but } \alpha x + (1-\alpha)y \neq z.$$

Therefore, the difference between strict convexity and convexity is that the former ~~is~~ for every bundle x and y weakly preferred to z , their convex combination is strictly preferred to z . But, the convex combination of x and y will be weakly preferred to z for convexity alone.